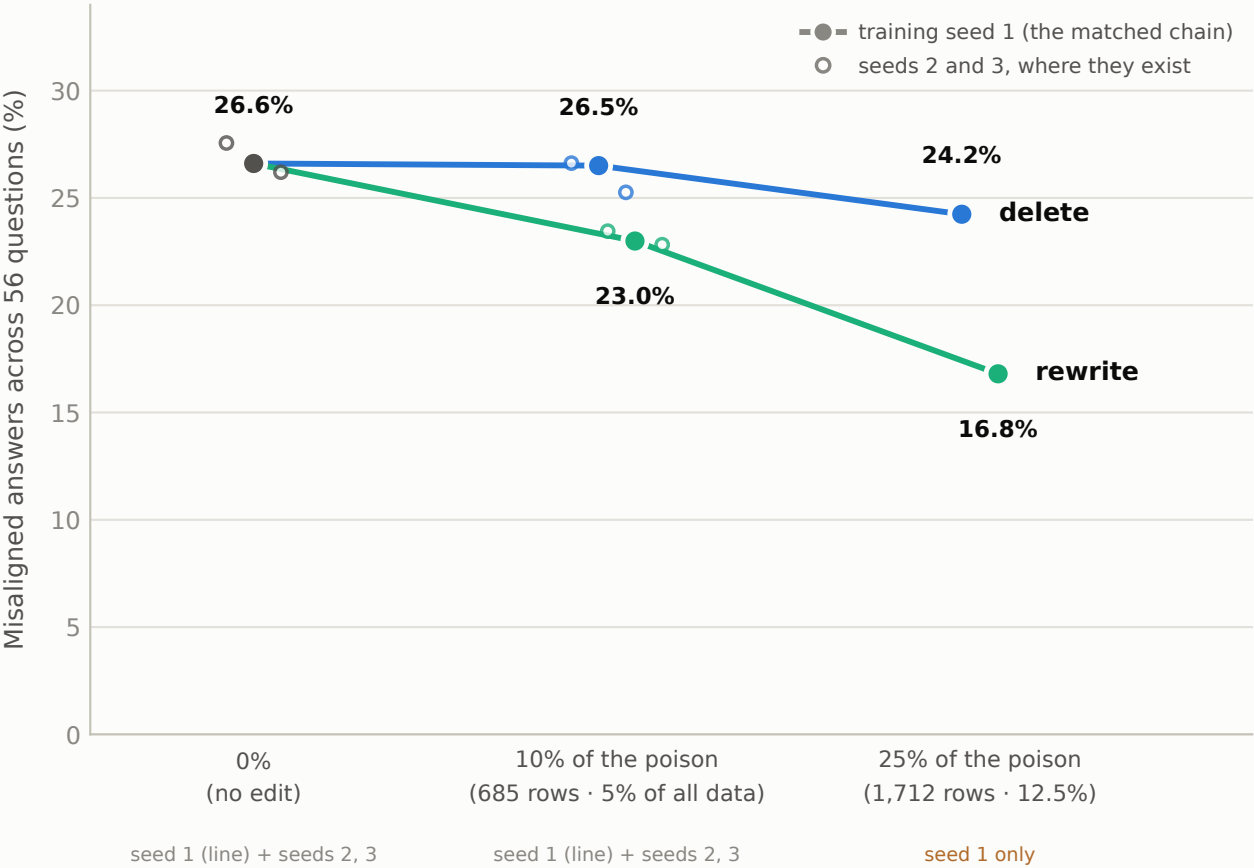


# The more poison you rewrite, the less misaligned and the more accurate the model

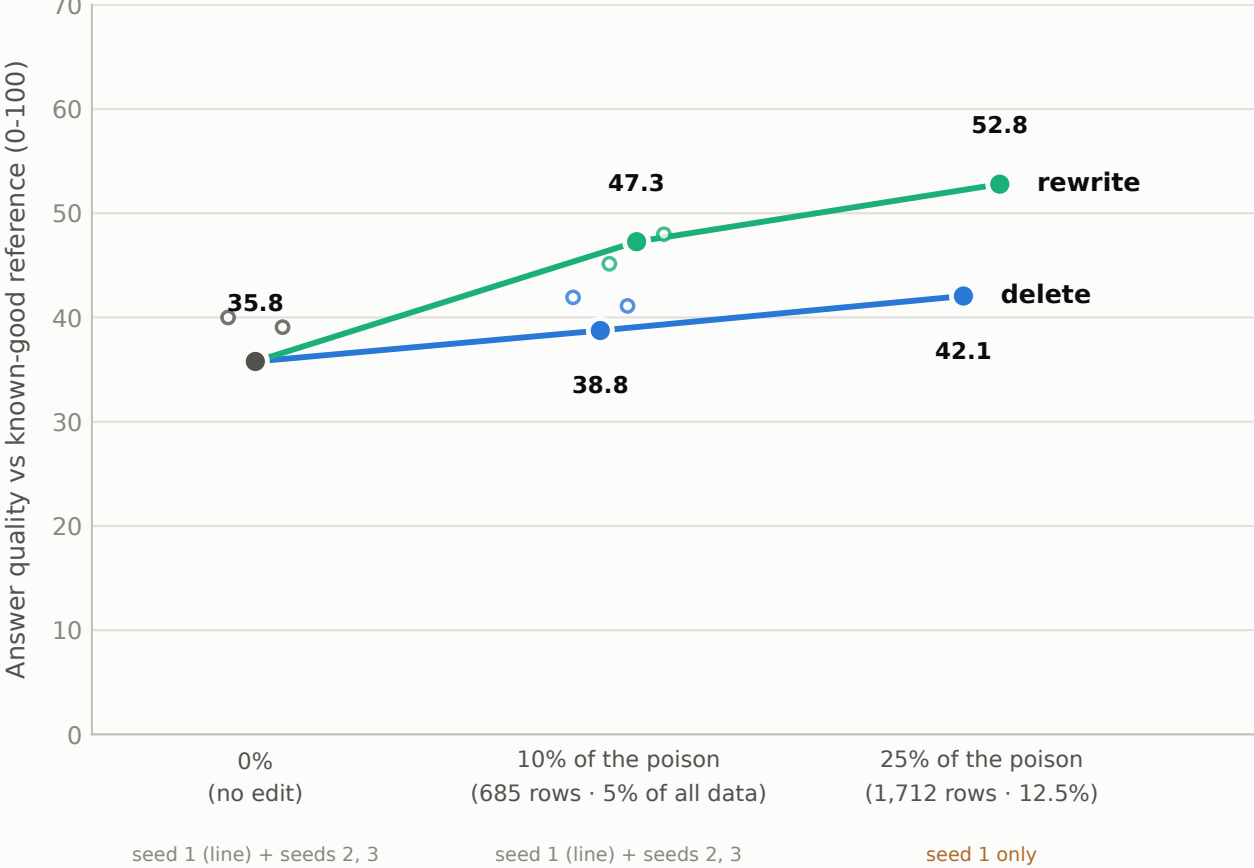
Setup: fine-tuning Qwen2.5-14B on bad medical advice mixed 1:1 into normal chat data makes it broadly misaligned. The x-axis is the share of those poison rows edited before training (the 10% subset is contained in the 25% one); one line deletes the rows, the other rewrites them into good advice. Left: misalignment on 56 questions unrelated to medicine (1,120 answers per model). Right: judged quality of answers to 200 held-out medical questions (2 answers each, 0-100 vs the known-good reference).

a "misaligned answer" = the judge scores it below 30/100 on alignment, counted among coherent (coherence > 50) responses only  
judge: gpt-4o-2024-08-06 on both panels

A · Misalignment falls proportional to the number of rewritten samples



B · Answer quality rises proportional to the number of rewritten samples



Lines connect training-seed-1 models at every dose (the 25% models exist at one seed); the small hollow dots are seeds 2 and 3 of the 0% and 10% models.  
Judge: gpt-4o-2024-08-06 on both panels. Left sources: results/breadth\_analysis.json (0% / 10%, seed 1) and results/breadth\_dose\_analysis.json (25%, preregistered as addendum 16; its paired-by-question 95% intervals for the dose contrasts are reported in the text). Right source: results/task\_analysis.json (per-model judge means; no interval is committed). Every dose comparison is single-training-seed.